VINAY SANKAR

GAME DEVELOPER

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Portfolio: terryberry10.github.io/portfolio

PROFESSIONAL SUMMARY

Game Developer with a solid foundation in Computer Applications (BCA) and advanced post-graduate specialization in Game Technology (M.Sc.). Skilled in Unity (URP), C# scripting, Shader Graph, HLSL shader programming, real-time particle VFX, and Mecanim animation state machines. Experienced in bridging art and engineering to craft responsive combat feedback, custom stylized shaders, and optimized game systems. Based in Bangalore, eager to contribute to mobile and PC game production pipelines.

TECHNICAL SKILLS & TOOLS

Game Engines: Unity (Advanced - URP / 2D / 3D), Unreal Engine, Godot
Shaders & VFX: Unity Shader Graph, HLSL Programming, Shuriken Particles, Combat VFX
Optimization & Profiling: Unity Profiler, Frame Debugger, Object Pooling, Overdraw Control
Programming & Scripting: C#, C++, C, OOP, JavaScript, HTML5
Animation & Rigging: Unity Mecanim Animator, Blend Trees, Ragdoll Physics, Maya / Blender
Tools & Version Control: Git, GitHub, Autodesk Maya, Blender

FEATURED GAME PROJECTS

Dottie — 2D Anime-Style 1v1 Combat Game (Unity / C#) — **Solo Developer**

- **Custom Shaders & Visual Juice:** Engineered stylized custom shaders in Shader Graph & HLSL for dynamic combat auras, afterimage motion trails, and distortion effects inspired by anime fighters.
- **Combat Feel & Game Feedback:** Designed responsive combo attack choreography, special attacks, and punchy audiovisual feedback loops (screen shake, hitstop freeze-frames, hit sparks, physics knockback).
- **Performance Optimization:** Built a modular C# Object Pooling architecture for high-frequency hit particles and projectiles, eliminating garbage collection spikes and ensuring smooth 60 FPS gameplay.
- **Full-Cycle Execution:** Solo developed all gameplay programming, animation states, custom shaders, and particle systems from scratch.

Astral Rally — Fast-Paced Multi-Ball Arcade Game (Unity / C#) — **Gameplay Programmer**

- **Shader & Arena Visuals:** Authored dynamic glowing neon shaders and reactive boundary collision materials that respond visually to high-velocity ball impacts.
- **VFX & Motion Trails:** Created procedural particle bursts and custom trail renderers to ensure high visual clarity during chaotic multi-ball gameplay.
- **Physics & Mechanics:** Implemented responsive paddle controls, multi-ball trajectory math, and collision detection in C#.

Bait — Reverse-Snake Survival Game (Unity / C#) — **Programmer**

- **Visual Indicators & Shaders:** Built custom trail shaders and danger-zone visual cues highlighting predator snake proximity.
- **Particle Feedback & Feel:** Designed lightweight particle burst systems triggered upon near-misses and capture events.
- **AI & Movement Logic:** Programmed predator pursuit algorithms and fluid player navigation mechanics in Unity.

FREELANCE EXPERIENCE

Freelance Web & Application Developer — Bangalore, India

- Built and deployed custom web applications and interactive interfaces using JavaScript and HTML5.
- Translated client specifications into clean, modular code with responsive UI and optimized assets.

EDUCATION

M.Sc. in Game Technology | ICAT Design & Media College (Alagappa University) — 2nd Year (Pursuing) • Chennai, India

Bachelor of Computer Applications (BCA) | Lovely Professional University (LPU) — Graduated 2025 • Punjab, India